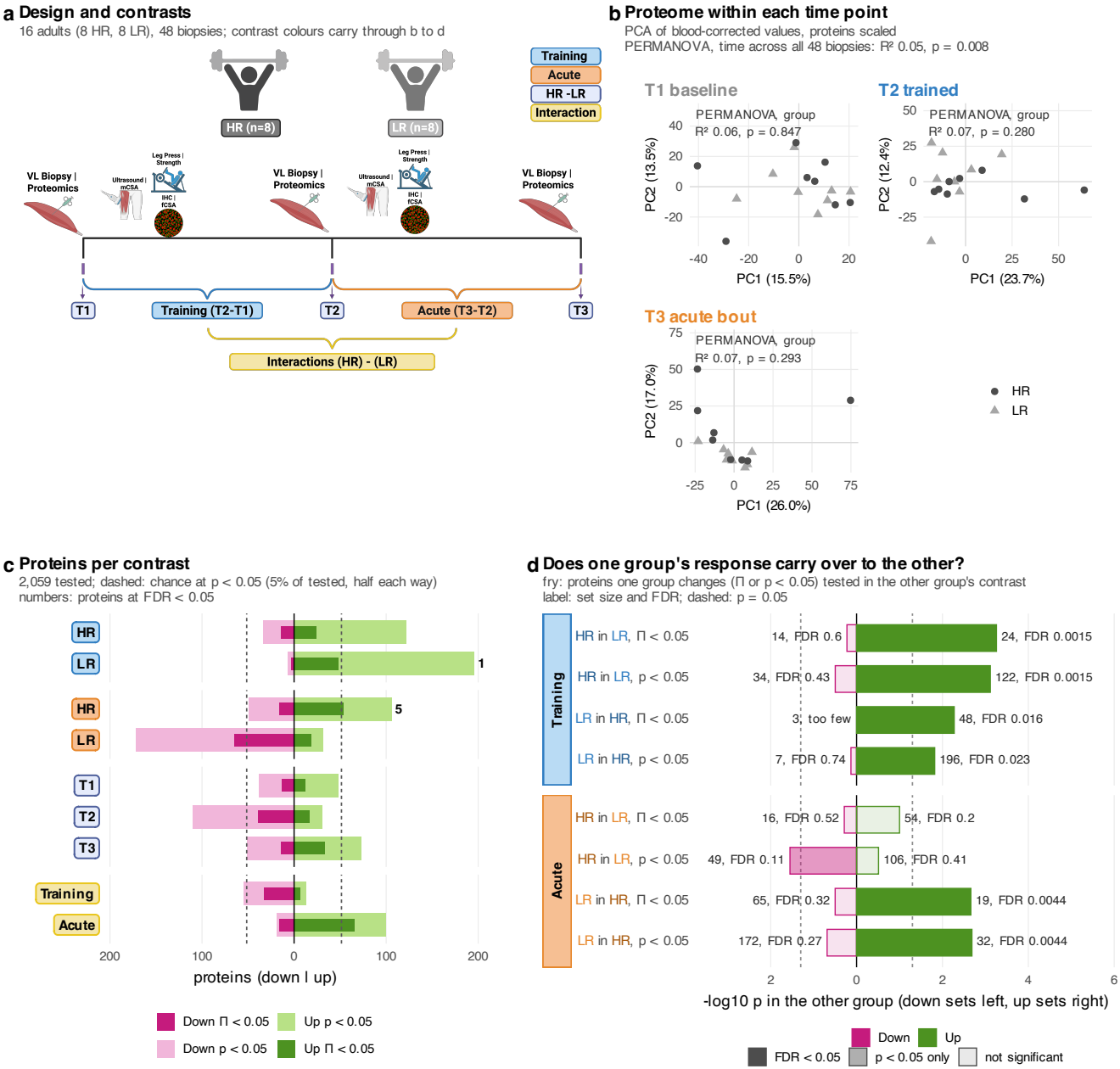

a Design and contrasts
16 adults (8 HR, 8 LR), 48 biopsies; contrast colours carry through b to d
HR (n=8)
LR (n=8)
Training
Acute
HR -LR
Interaction
VL Biopsy | Proteomics
Ultrasound | mCSA
Leg Press | Strength
IHC | fCSA
T1
Training (T2-T1)
T2
Acute (T3-T2)
T3
Interactions (HR) - (LR)
b Proteome within each time point
PCA of blood-corrected values, proteins scaled
PERMANOVA, time across all 48 biopsies: R² 0.05, p = 0.008
T1 baseline
PERMANOVA, group
R² 0.06, p = 0.847
T2 trained
PERMANOVA, group
R² 0.07, p = 0.280
T3 acute bout
PERMANOVA, group
R² 0.07, p = 0.293
PC1 (15.5%)
PC2 (13.5%)
PC1 (23.7%)
PC2 (12.4%)
PC1 (26.0%)
PC2 (17.0%)
HR
LR
c Proteins per contrast
2,059 tested; dashed: chance at p < 0.05 (5% of tested, half each way)
numbers: proteins at FDR < 0.05
HR
LR
T1
T2
T3
Training
Acute
1
5
proteins (down | up)
Down Π < 0.05
Up p < 0.05
Down p < 0.05
Up Π < 0.05
d Does one group's response carry over to the other?
fry: proteins one group changes (Π or p < 0.05) tested in the other group's contrast
label: set size and FDR; dashed: p = 0.05
Training
Acute
HR in LR, Π < 0.05
HR in LR, p < 0.05
LR in HR, Π < 0.05
LR in HR, p < 0.05
14, FDR 0.6
24, FDR 0.0015
34, FDR 0.43
122, FDR 0.0015
3, too few
48, FDR 0.016
7, FDR 0.74
196, FDR 0.023
16, FDR 0.52
54, FDR 0.2
49, FDR 0.11
106, FDR 0.41
65, FDR 0.32
19, FDR 0.0044
172, FDR 0.27
32, FDR 0.0044
-log10 p in the other group (down sets left, up sets right)
Down
Up
FDR < 0.05
p < 0.05 only
not significant